

# EphemeralTrust: Session State and Observation Design in Controlled AWS Access Traces

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## Abstract

A cloud configuration can deny new role sessions while a previously issued credential still permits a resource request. Reconstructing that access requires observing both configuration and issued-session state, but the resulting scores also depend on observation timing and witness matching. We examine these effects in six controlled AWS traces connecting GitHub OpenID Connect, the Security Token Service and an S3 endpoint. In S3, one issued session completed 168 successful access probes paired with denied fresh issuance. At 60-second sampling, configuration-only B1 recovered the selected witness's semantic tuple in 0/10 phase projections, while session-aware B2 recovered it in 10/10 under the same configuration schedule. Coarser B1 observations scored better because stale issuance paths matched the tuple without representing the observed session mechanism. Denser observations also improved first-witness scores; the same-input T-S control is structurally identical to B2. In S5, atemporal aggregation produced a candidate incompatible with the declared observed credential schedule. A separate offline reconstruction reproduced 66 analysis files and matched 1,200 witness-state decisions. The results distinguish semantic witness recovery from faithful reconstruction of the credential mechanism and show how observation density affects the comparison.

Keywords: AWS IAM; temporary credentials; OIDC; session state; observation design; access reconstruction

## 1 Introduction

An IAM trust policy can stop admitting new role sessions while a credential obtained earlier remains usable for a resource request. A configuration readback after that change describes the issuance relationship, but historical access reconstruction also needs the credentials already obtained. Removing an issuance edge from a model can otherwise discard state needed to explain an observed request.

This paper uses session persistence in a narrow empirical sense: sampled successful use of an issued session while fresh issuance is denied. The distinction is visible in our S3 scenario, where one session succeeded in 168 paired observation cycles. Periodically sampled configuration-only B1 missed the selected witness's semantic tuple at every 60-second phase, while session-aware B2 recovered it under the same configuration schedule.

The score itself gave a less obvious result. B1's S3 score increased as sampling became coarser, from 0/10 at 60 seconds to 10/10 at 900 and 1,800 seconds. Stale observations still described an open fresh-issuance path that matched the same context, roles, action and resource as the old-session request. The semantic score improved even though the candidate represented the observed mechanism less faithfully. In another scenario, periodic observations captured a temporary capability later in its episode but missed its first successful witness.

Prior work models acquired credentials and policy-changing AWS actions [1], and reconstructs observed role-assumption chains using temporal session graphs [2]. Our research question concerns measurement: In controlled AWS OIDC-to-STs access traces, how do observed session persistence, configuration sampling and witness-selection rules affect recovery and interpretation of ephemeral privilege paths? A secondary question asks when an atemporal union of configuration and session evidence produces a path incompatible with the declared credential schedule.

The six traces link configuration changes and issued credentials to direct resource-use witnesses. The accompanying analysis is reproducible offline.

## 2 Background and related work

### 2.1 Credential issuance and use

GitHub Actions can obtain an OIDC JSON Web Token when a workflow has the relevant token permission. The setting `id-token: write` permits token retrieval, not AWS resource access [3]. The workflow submits that token to `AssumeRoleWithWebIdentity`, which returns AWS temporary credentials [4]. Those credentials can support authorized resource requests and, where permitted, a subsequent `AssumeRole` operation [5]. The OIDC token, entry-role credentials and child-role credentials are distinct objects in the experiment.

AWS separates the trust relationship governing role assumption from the permissions governing role operations [5]. Temporary credentials do not preserve a fixed copy of resource permissions: applicable permissions are evaluated when requests are made, and changes can affect credentials issued earlier [6]. AWS also documents policy-based revocation of older role sessions [7]. Blocking fresh issuance, removing a resource permission and reaching credential expiry are consequently different interventions.

Returned expiration bounds a credential's lifetime but does not guarantee authorized use throughout that lifetime [8]. Our reconstruction uses observed issuance and expiration values, configuration readbacks and direct requests. Since IAM changes are not always immediately visible [9], the recorder retains request start and completion times. Administrative acknowledgment, observed configuration and resource behavior are recorded separately.

### 2.2 IAM reasoning and cloud identity history

Zelkova uses SMT-based reasoning for AWS access policies [10], while Barnett et al. model the AWS authorization engine more broadly [11]. These works provide policy-semantics foundations beyond the subset exercised here. Shevrin and Margalit's model checking includes acquired credentials and policy-changing actions [1]. Its abstraction excludes expiration during the attack and ignores `DurationSeconds`; our experiment measures particular credential lifetimes under a fixed schedule. The CloudLens preprint models cloud vulnerabilities through relational planning and permission-flow actions, including role assumption [12].

RoleSentry represents AWS identities, assumed-role sessions and temporal role-assumption relationships from observed activity [2]. Its session representation uses role ARN and role-session name. EphemeralTrust records credential-instance lineage alongside controlled configuration changes and verified resource-use witnesses. Parisel's temporal fiber model studies evolving non-human identity permission profiles in Azure [13]. Its profile lineage concerns changing permission groups, distinct from STS parent-child issuance.

Our experiment draws on this work on policy reasoning and identity history. The comparison here is among representation and observation conditions within one implementation.

### 2.3 Delegation, temporal order and sampling

UCON models authorization before and during use [14]; Macaroons support constrained delegation through contextual caveats [15]; and WAVE addresses transitive delegation with its own proof and lifecycle mechanisms [16]. They provide related treatments of continuing authority and delegation, although their mechanisms differ from STS.

Temporal-network research explains why aggregating edges can discard the order required for a time-respecting path [17], and how sampling affects inferred temporal properties [18]. AWS credentials add a concrete lifecycle: successful issuance creates state that can remain relevant after the issuing configuration changes. Our traces examine that lifecycle under controlled sampling and a first-success witness rule.

## 3 Experimental design

### 3.1 System and observations

The system uses one AWS account in `us-east-1`, a shared S3 canary bucket, and per-run entry and chain roles with distinct object prefixes. The controller installs scheduled IAM changes. The experiment actor obtains

credentials and attempts the specified resource operation. An S5 verifier uses separate authority to check the later resource capability, while S6 includes a main-context companion in a separate workflow job.

Each successful issuance creates a ledger entry with identity, request timing, returned expiration and exact parent association where applicable. Resource probes use existing credentials and verify the payload or ETag as appropriate; they do not issue sessions. Configuration readbacks describe the modeled trust and resource-permission state.

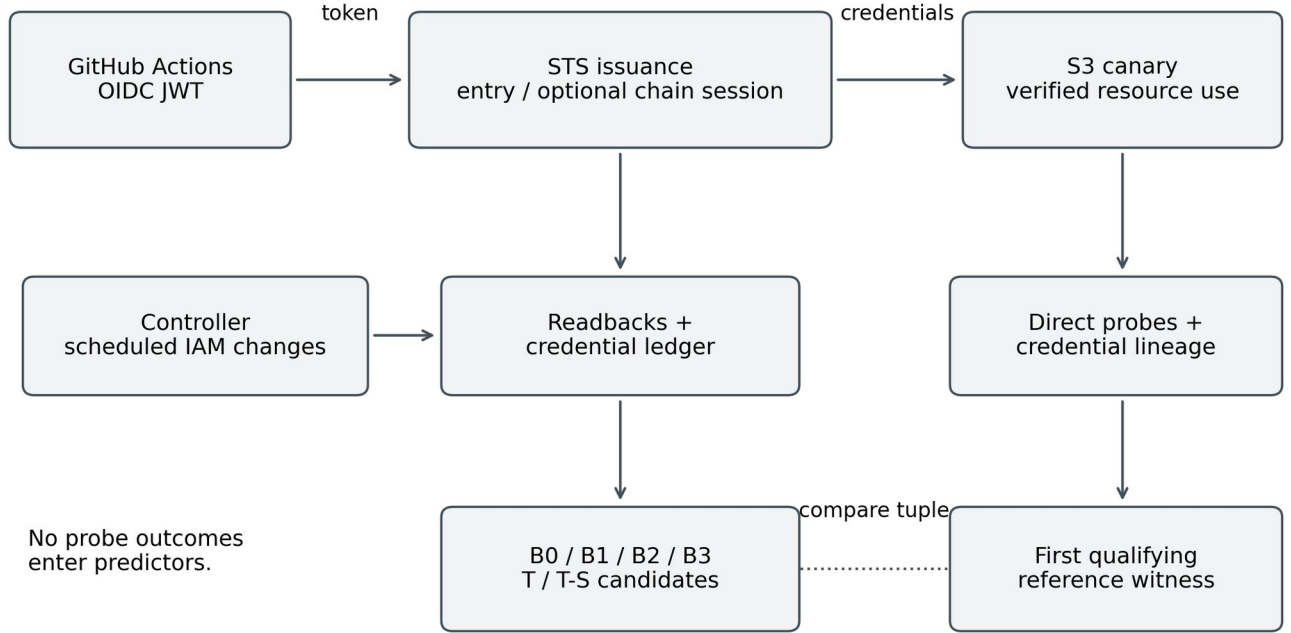


Figure 1. Credential and evidence flow. Predictors read configuration and session records; the reference selects verified direct outcomes with lineage checks. The dotted line denotes semantic comparison. Recording infrastructure and some lineage helpers are shared.

Predictors read configuration and credential records through a restricted observation object. A separate reference builder selects verified direct access witnesses and their lineage without invoking the candidate predictor. Probe outcomes do not enter predictors. The two paths share recording infrastructure, timestamps and some lineage and semantic helpers. A separate offline reconstruction checks the observed cases without calling the project's IAM/path predictor.

CloudTrail provides corroborating management-event evidence linked by recorded request identifiers. Direct probes supply the resource outcomes because this collection contains no S3 data events. AWS distinguishes management and data events and does not guarantee event-file ordering [19].

### 3.2 Dataset and scenarios

The dataset consists of six audited runs, one per scenario. Each has a nominal duration of 7,200 seconds and intended 10-second polling. Table 1 lists the interventions. S1 provides a stable control; S2 changes a resource grant; S3 separates issuance from existing-session use; S4 adds and removes a role chain and its resource capability; S5 separates those opportunities with an expiry barrier; and S6 changes the admitted OIDC context while the main-context companion samples access.

Table 1. Scenario design. Times are scheduled seconds from trace start; each run lasts nominally 7,200 s.

| ID | Intervention                     | Schedule (s)                                                               | Purpose                                |
|----|----------------------------------|----------------------------------------------------------------------------|----------------------------------------|
| S1 | Stable GetObject control         | No scheduled mutation                                                      | Stable sampled operation               |
| S2 | Entry-role PutObject grant       | Grant 1,800; revoke 2,400                                                  | Early witness versus later observation |
| S3 | Old session and blocked issuance | Issue 1,740; block 1,800; restore 3,600                                    | Issuance/use separation                |
| S4 | Entry → chain → canary           | Chain 1,800; capability 2,400; remove capability 3,000; remove chain 3,300 | Sequential grants and removals         |
| S5 | Separated chain and capability   | Chain 1,200–1,800; capability 3,600–4,200; expiry barrier                  | Declared-schedule incompatibility      |
| S6 | Alternate OIDC context           | Broaden 1,800; restore 2,400                                               | Context boundary with main control     |

Only these six runs enter the reported results. Earlier attempts and their exclusions are documented in the supplement.

### 3.3 Scope of the reconstruction

The evaluator supports the experiment's four actions: `sts:AssumeRoleWithWebIdentity`, `sts:AssumeRole`, `s3:GetObject` and `s3:PutObject`. Its condition subset covers `StringEquals`, `StringLike` and `ArnEquals` over supported OIDC audience/subject and principal-ARN keys. The live policies use generated Allow statements, constrained subjects and exact resource scope. Chaining uses same-account root delegation constrained by the entry-role ARN. AWS policy composition is broader than this subset [20].

Unsupported syntax, malformed observations and unsupported scope yield UNKNOWN. The live interventions remove grants or change trust. A denied request does not by itself identify an explicit policy Deny.

An observed session becomes a starting state only for the relevant experiment actor and context. Its lineage must connect the recorded parent and child, with parent validity checked at child issuance. At witness time the session must be visible to the observer and within its own recorded lifetime. Configuration edges then determine continuation to the protected endpoint. Reaching an intermediate role alone is insufficient.

For S5, the certificate's credential universe is the experimental actor's complete issuance ledger under the declared schedule, assuming no undeclared credentials or writers. The verifier's credentials are outside that universe. The certificate evaluates whether the observed credentials can connect the separated phases.

### 3.4 Representation and observation conditions

Periodic projections use intervals of 60, 300, 900 and 1,800 seconds, each with ten offsets  $jS/10$  for  $j = 0, \dots, 9$ . At a nominal sample, the latest completed readback is selected only if no more than 15 seconds old. An incomplete latest readback is not replaced with an older successful one. Eligibility requires four interval spans and complete scheduled coverage.

The conditions differ in the information they use and when they observe it. B0 uses the final configuration without session starts and is retrospective. B1 uses periodically sampled configuration, held until the next sample. B2 adds observed sessions available by that nominal sample, including lineage and expiry. B3 forms an atemporal union of sampled configuration edges and recorded session starts, discarding ordering and expiry for candidate construction.

T uses the same candidate logic as B2 with denser readbacks and the issuance ledger available by witness start. T-S supplies B2's inputs and contract and is structurally identical to B2. Differences involving T therefore measure an observation condition and do not establish algorithmic superiority.

B3 has access to the ledger through trace end, including issuance after the final nominal sample, giving it a larger retrospective information budget. Restricting that tail in the offline check changed none of the 240 B3 candidate cells. B0 and B3 are scored as retrospective hypotheses; B1, B2, T and T-S use witness-time sample-and-hold.

### 3.5 Semantic witness recovery and timing

The primary metric is semantic-tuple witness recovery, shortened below to semantic witness recovery. A matching tuple contains the workflow context, ordered roles, action and resource. It does not require identification of the exact credential that produced the request. Mechanism-consistent reconstruction additionally depends on the session and lineage evidence.

Each predefined positive episode contributes its first qualifying successful witness. S1 eligibility begins at 1,800 seconds, after initial sampling boundaries. S3 additionally requires fresh-issuance denial in the same cycle as old-session success. The reference checks payload verification, recorded lineage, episode containment and credential lifetime. There are five positive references, from S1, S2, S3, S4 and S6. S5 has a separate schedule-certificate task and no positive-reference recall denominator.

Each fraction reports how many eligible phase projections recovered the scenario's single reference witness. Historical structure recovery instead counts whether a matching tuple appeared anywhere in the candidate history.

Timing measurements retain request brackets. A transition is confirmed when three consecutive expected completions occur within 300 seconds of its administrative acknowledgment. Conditional window lengths additionally assume a single monotone transition into and out of the observed allowed episode. Full-precision values and exclusion mechanics are in the supplement.

## 4 Results

### 4.1 Existing-session access continued while fresh issuance was denied

S3's old session was issued at 1,740.46 seconds and had returned expiration at 3,540.38 seconds. After the scheduled trust change, the first observed fresh-issuance denial completed at approximately 1,862.29 seconds. The same cycle's old-session GetObject then succeeded. Across 168 paired sampled cycles, this one session continued accessing the object while fresh issuance was denied. Its final successful request ended at 3,532.63 seconds, before expiration.

Figure 2 plots the successful samples and the request brackets around the first pair. The requests directly distinguish the state needed to issue another credential from the state used by the existing credential.

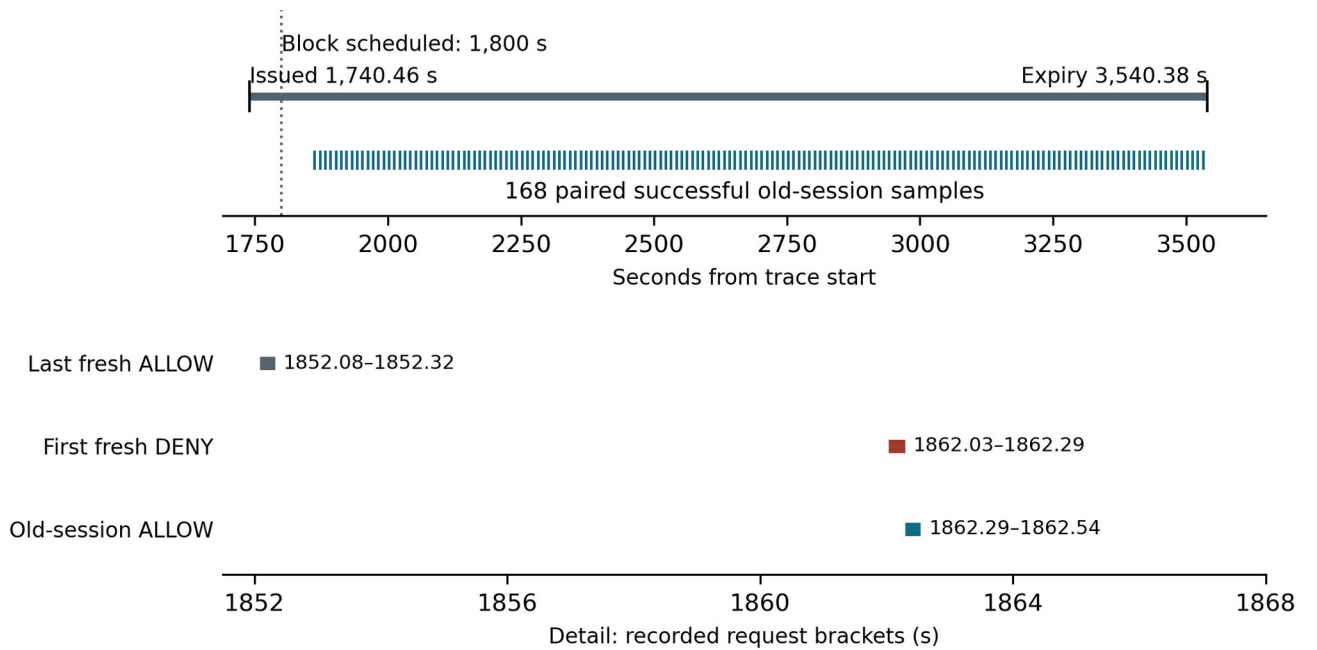


Figure 2. S3 session lifetime and paired samples. The gray line denotes credential lifetime; teal marks show 168 successful old-session completions paired with fresh-issuance denial. The lower panel shows request brackets near the first pair. Labels are rounded to 0.01 s; full-precision coordinates are preserved in the supplement. The intervention line marks scheduled time.

Under 60-second sampling, B1 recovered the selected witness's tuple in 0/10 projections and B2 in 10/10 (Table 2). The configuration schedule was identical. B2 could start from the recorded unexpired session after the fresh-issuance edge disappeared, so the represented session state accounts for the difference.

#### 4.2 Coarser observations received better semantic scores

B1's S3 score rose to 8/10 at 300 seconds and 10/10 at 900 and 1,800 seconds. The coarser observations still showed an open fresh-issuance path. That stale path had the same context, role sequence, action and resource as the old-session reference, so it received matching credit without representing the observed persistence mechanism.

Finer configuration-only observations correctly reflected removal of the issuance edge, but lacked the session state needed to explain continued use. Coarser observations retained an obsolete issuance path and scored better. The semantic score alone cannot distinguish those explanations.

B0 also matched the S3 tuple using the restored final configuration. It matched structure from outside the blocking period. The controlled comparison is between periodically sampled B1 and session-aware B2.

Table 2. Selected semantic witness recovery results. Fractions count recovered / eligible phase projections. T uses denser evidence; B2 and T-S are structurally identical. S4 uses nine eligible phases. Complete results, including B0 and B3, are in Table S1.

| Scenario | Interval (s) | B1    | B2 / T-S | T     | Measurement effect             |
|----------|--------------|-------|----------|-------|--------------------------------|
| S3       | 60           | 0/10  | 10/10    | 10/10 | Observed session state         |
| S3       | 300          | 8/10  | 10/10    | 10/10 | Stale issuance-path match      |
| S3       | 900          | 10/10 | 10/10    | 10/10 | Stale issuance-path match      |
| S2       | 60           | 1/10  | 1/10     | 10/10 | Early-witness observation      |
| S2       | 300          | 0/10  | 0/10     | 10/10 | Later structure still observed |
| S4       | 60           | 2/9   | 2/9      | 9/9   | One phase excluded             |
| S6       | 60           | 2/10  | 2/10     | 10/10 | Early-witness observation      |

#### 4.3 First-witness selection favored denser observations

S2 recorded 60 successful PutObject probes and 660 denials, with successful fresh issuance throughout the sampled trace. Its first successful resource request completed at approximately 1,811.68 seconds. B1, B2 and T-S recovered that witness in 1/10 phases at 60 seconds and 0/10 at longer intervals. Their historical structure recovery, however, was 10/10 at both 60 and 300 seconds. The periodic observations captured the granted capability later in the episode.

T recovered the selected witness in all ten phases at each interval. The recorder takes configuration readbacks before issuance and canary work in a tick, giving the denser condition evidence close to an early successful request. When T-S receives B2's information and scoring contract, it reproduces B2's result.

Assuming a single monotone allowed episode, S2's allowed-use window lasted approximately 589.83–609.69 seconds. A uniformly phased periodic sample hits a window of length  $L$  with idealized probability  $\min(1, L/S)$ , where  $S$  is the interval. At 900 seconds, the conditional bounds are approximately 0.655–0.677. The ten deterministic phase fractions are 1, 1, 0.6 and 0.3 across the four intervals.

A sample can hit this window after the first successful witness, explaining the poor first-witness scores. S6's fresh-issuance window, under the same assumption, lasted approximately 629.83–649.92 seconds, with phase fractions of 1, 1, 0.7 and 0.3. These bounds describe sampling geometry, not confidence intervals.

#### 4.4 Atemporal aggregation violated the observed credential schedule

S5 separated the opportunity to obtain a child session from a later resource grant. The experimental actor obtained 60 child sessions in the earlier phase. The latest expiration was at 2,702.14 seconds, approximately 897.86

seconds before the capability phase scheduled at 3,600 seconds. This exceeded the required 30-second expiry margin. None of those recorded child credentials could bridge the two phases.

The experimental actor recorded 148 canary denials and no success. The separate verifier recorded 60 successes and 30 denials, providing sampled evidence that the later capability operated under verifier authority. All eight certificate checks passed under the declared ledger and writer assumptions.

B3 nevertheless emitted the schedule-incompatible union candidate in 10/10, 10/10, 6/10 and 3/10 phases at 60, 300, 900 and 1,800 seconds. It combined observed ingredients from different times after discarding their ordering and expiry relationship. The other conditions emitted no certified invalid candidate in these cells.

The certificate establishes incompatibility within the declared observed credential universe, not universal AWS nonreachability.

#### 4.5 Boundary scenarios and verification

Table 3 summarizes the direct outcomes. S1 recorded 720 fresh-issuance successes and 720 verified GetObject successes, with semantic witness recovery in every phase at every interval. S4 recorded successful chained access, followed by resource denials after capability removal and child-issuance denials after chaining removal. This complements S3 by distinguishing changes to resource use from changes to issuance.

Table 3. Direct observations. Counts are within-run records. S6 control success is sampled; the S5 verifier is a separate actor.

| ID | Direct outcomes                                                                  | Scope                                      |
|----|----------------------------------------------------------------------------------|--------------------------------------------|
| S1 | 720 fresh issuance successes; 720 verified GetObject successes                   | One stable trace                           |
| S2 | 60 PutObject successes; 660 denials                                              | Grant/remove response                      |
| S3 | 168 paired old-session successes and fresh-issuance denials                      | One observed old session                   |
| S4 | 150 child issuances; 60 canary successes; one readback timeout                   | Six rows excluded; 234/240 eligible        |
| S5 | 60 child issuances; 148 experimental canary denials; zero experimental successes | Verifier: 60 successes, 30 denials         |
| S6 | 181 → 64 → 475 fresh-issuance DENY/ALLOW/DENY counts; 64 canary successes        | Main companion: 720/720 in-trace successes |

One S4 readback timeout was marked UNKNOWN, not counted as an authorization denial. The coverage rule excluded six condition rows, leaving 234/240 eligible and nine phases for the 60-second fractions in Table 2. The complete-interval aggregator excludes that interval altogether. Supplement S3 gives the timing and phase identifiers.

In S6, alternate fresh issuance followed 181 denials, 64 successes and 475 further denials; all 64 associated canary probes succeeded. The main-context companion recorded 720 successful in-trace completions, with a maximum gap of approximately 10.42 seconds. This establishes sampled main-context operation during the intervention, not continuous availability. The alternate runner did not test old credentials after fresh issuance returned to denial.

All 14 transition series in S2–S6 met the three-completion confirmation rule. An offline reconstruction reproduced 66 analysis files byte-for-byte and independently matched 1,200 witness-state decisions using a separate narrow evaluator. These checks concern the recorded cases. Checksummed source, evidence and reproduction details are documented in the supplement.

## 5 Discussion

### 5.1 Representing the credential changes the reconstruction

After trust blocking in S3, configuration-only B1 represented the absence of a fresh-issuance path but could not use an earlier credential as its starting state. B2 supplied that state and recovered the selected tuple. The paired requests show that the earlier credential was still being used.

The contents of the snapshot also matter here. A snapshot containing the issued credential, its lifetime and sufficient resource-permission information can represent the observed capability. S3 therefore gives a specific reason to preserve credential-instance evidence; it does not establish a general limitation of snapshot analysis.

S4 helps interpret what that credential evidence means. Its resource denials after capability removal contrast with S3's success after issuance blocking. Expiration bounds the ledger's usable starting states, while resource-permission state determines their continuation. Possessing an unexpired credential alone is insufficient to explain successful access.

## 5.2 The score can improve while the explanation worsens

In coarse S3 projections, the semantic tuple matched even though the candidate relied on stale fresh-issuance state. The recovery fraction alone hides why the score improved: it was still using an obsolete issuance path. The paired requests make this discrepancy visible because they identify the existing session's use alongside denied issuance.

S2 exposes a different scoring choice. Selecting the first success emphasizes the onset of an episode, so periodic conditions can capture the capability later and still score poorly. Historical structure recovery records that later evidence. Both measurements are useful, but they answer different questions about the same trace.

The denser T condition had evidence closer to the first request, while T-S controlled for that information difference. A reconstruction comparison therefore needs to report the observation budget and matching identity alongside its score.

## 5.3 A union candidate needs a schedule interpretation

B3 found the ingredients of an S5 path in the collected evidence. The expiry barrier prevented the recorded child sessions from connecting those ingredients in the declared order. Aggregating the states removed the constraint that mattered in this scenario.

The certificate adds evidence beyond the zero-success probe count: recorded lineage and expiration constrain the available credentials, phase separation defines the barrier, and verifier success confirms sampled operation of the later capability. Without those records, the denied requests alone would provide a much weaker explanation of why the experimental actor could not connect the phases.

## 6 Limitations

The study covers six controlled scenarios, with one analyzed run per scenario in one AWS account and region. Phase projections, probes, credential records and reconstructed decisions are dependent within-run observations, not experimental replicates. The findings are descriptive rather than population-level estimates, and the research question was refined after examining the evidence. The evaluator covers the restricted IAM subset defined in Section 3.3, not general AWS authorization. The supplement lists the untested policy mechanisms. Some lifecycle cases, including isolated child use after parent expiry, have only offline validation.

Measurement depends on positive first-success references and semantic matching, which does not identify the exact credential responsible for a request. Predictors and references share recording and lineage infrastructure, despite the separate offline implementation check. S5's conclusion depends on its declared credential-universe and writer assumptions. Sequential readbacks and request brackets do not identify exact AWS enforcement instants. The S4 coverage exclusion is retained without assuming its missing observation was random.

## 7 Conclusion

In S3, one issued AWS session continued accessing the object in 168 sampled cycles while fresh issuance was denied. Adding that session state changed semantic witness recovery from 0/10 to 10/10 under the same 60-second configuration schedule. Coarser configuration-only observations sometimes scored better because stale issuance state matched the tuple, while denser observations recovered an early witness that periodic observations captured only later.

S5 showed a different consequence of losing temporal information: atemporal aggregation combined observed states into a candidate incompatible with the declared credential schedule. These traces show why a recovery score needs to be read alongside the represented session state, observation schedule and witness definition.

## Artifact availability

The accompanying supplement contains complete result tables, full-precision timing, source and archive identifiers, attempt provenance, exclusions and reproduction checks. The package's evidence map identifies the underlying archived traces and implementation. A separate verification record maps numerical claims and citations to their sources. The artifacts are supplied with the manuscript.

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